Worksheet 1

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1.1 Aim/Overview of the practical:

To understand the basics of Python and its importance in Machine Learning.

1. To study and implement basic Python scientific libraries:

Math

NumPy

Matplotlib

Seaborn

SciPy

Procedure:

1. Install Python from the official Python website.
2. Create a virtual environment (optional).
3. Install required packages using pip.
4. Execute Python code to test each library.
5. Verify outputs and plots.

Program / Code:

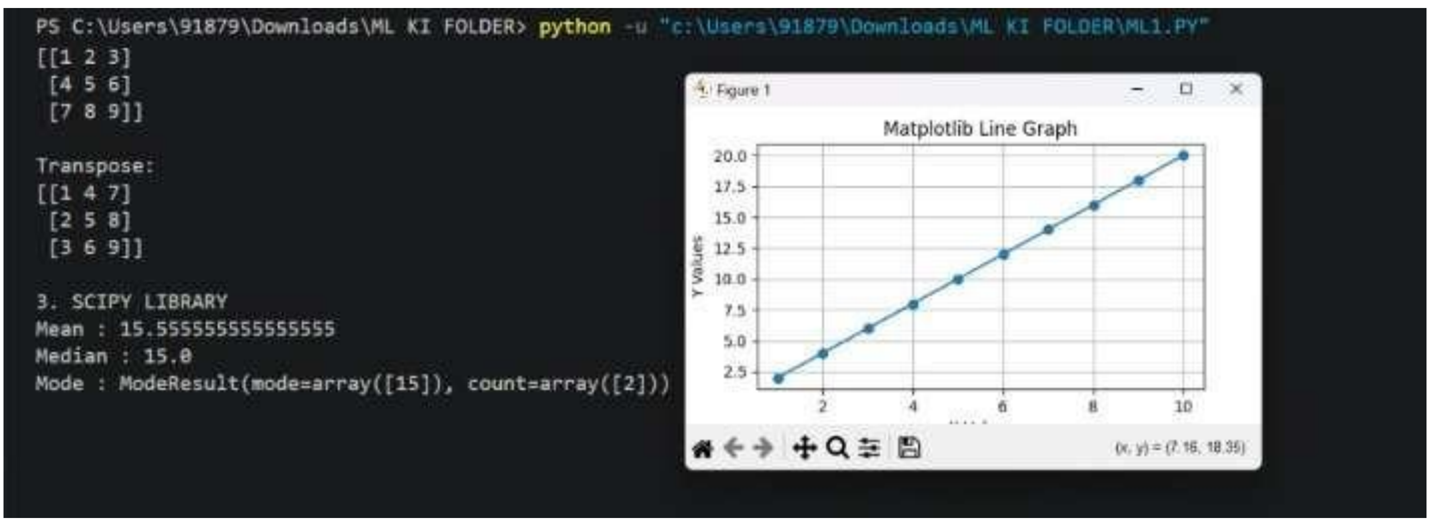
```
# Python Scientific Libraries Demo

# Math | NumPy | Matplotlib | Seaborn | SciPy

import math import numpy as np

import matplotlib.pyplot as plt
```

```
import seaborn as sns

from scipy import stats

# 1. Math Library

print("1. MATH LIBRARY")

number = 25

print("Square Root :", math.sqrt(number))

print("Factorial   :", math.factorial(5))

print("Power       :", math.pow(3, 4))

print("Value of Pi :", math.pi)

print("Sin(90°)    :", math.sin(math.radians(90)))

print()

# 2. NumPy Library

print("2. NUMPY LIBRARY")

arr = np.array([10, 20, 30, 40, 50])

print("Array :", arr)

print("Mean  :", np.mean(arr))

print("Sum   :", np.sum(arr))

print("Max   :", np.max(arr))

print("Min   :", np.min(arr))

print("Std Dev :", np.std(arr))

matrix = np.array([[1,2,3],[4,5,6],[7,8,9]]
print("\nMatrix:")

print(matrix)
```

```
print("\nTranspose:")

print(matrix.T)

# 3. SciPy Library

print("\n3. SCIPY LIBRARY")
data = [12,15,14,10,18,20,15,17,19]
print("Mean :", np.mean(data))
print("Median :", np.median(data))
print("Mode :",stats.mode(data, keepdims=True))

 # 4. Matplotlib Graph

x = np.arange(1,11)
y = x * 2
plt.figure(figsize=(7,4))
plt.plot(x, y, marker='o')
plt.title("Matplotlib Line Graph")
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.grid(True) plt.show()

 # 5. Seaborn Graph

 plt.figure(figsize=(7,4))

 sns.barplot(x=["A","B","C","D","E"]

 ,   y=[12,25,18,30,20])

plt.title("Seaborn Bar Chart")

plt.show()

# 6. Histogram

marks =

[55,67,72,80,91,65,70,88,76,95,62,78,81,

69,73,84,90,66,79,85]
```

```
plt.figure(figsize=(7,4))

sns.histplot(marks, bins=6, kde=True)

plt.title("Student Marks Distribution")

plt.xlabel("Marks")

plt.ylabel("Frequency") plt.show()

 print("\nProgram Executed Successfully!")
```

Output:



Result:

Thus, the basic Python libraries Math, NumPy, Matplotlib, Seaborn, and SciPy were successfully installed and executed. The environment is now ready for Machine Learning and data analysis tasks.

Conclusion:

Python provides a powerful ecosystem for Machine Learning. By using scientific libraries, complex mathematical operations, data visualization, and optimization tasks can be performed efficiently. This experiment confirms the successful setup of the Python environment required for ML applications.